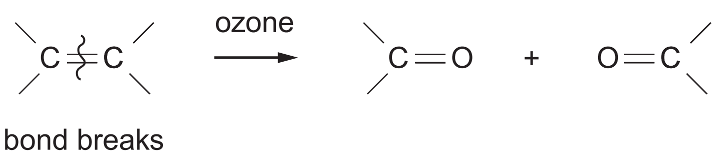
Examiner only

**9.** *(a)* Alkenes react with ozone to give an intermediate compound, which can then react further to give aldehydes or ketones.




1 mol of an alkene **T** reacts with ozone to give 2 mol of the same carbonyl compound, **U**.

(i) Explain what can be deduced about the structural formula of alkene **T**. [1]

.............................................................................................................................................

.............................................................................................................................................

(ii) Carbonyl compound **U** reacted with 2,4-dinitrophenylhydrazine to produce a solid derivative.

State the colour of this solid derivative. [1]

..............................................................

(iii) I. The $^{1}$H NMR spectrum of compound **U** did **not** show a chemical shift at 9.8 δ.

State what can be deduced from this statement. [1]

.............................................................................................................................................

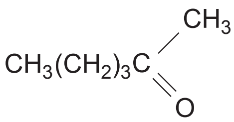
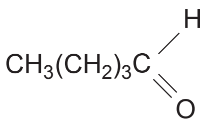
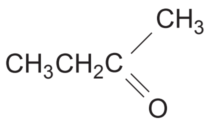
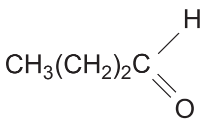
.............................................................................................................................................



Examiner only

II. The melting temperatures of the 2,4-dinitrophenylhydrazine derivatives can be used to identify the original carbonyl compound. The melting temperatures of some of these derivatives are shown in the table.

| Carbonyl compound | Melting temperature of derivative / °C |
|---|---|
|  | 106 |
|  | 108 |
|  | 115 |
|  | 126 |

1410U401 09

The 2,4-dinitrophenylhydrazine derivative of compound **U** was slightly impure and melted at a temperature of 110-113 °C.

Use the information given to deduce the formula of compound **U**.
Explain your reasoning. [3]

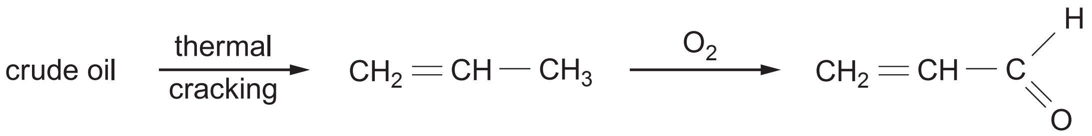
(iv) Use your answer to part (iii) and other relevant information given in the question to give the structure of alkene **T**. Name alkene **T**. [2]

Name

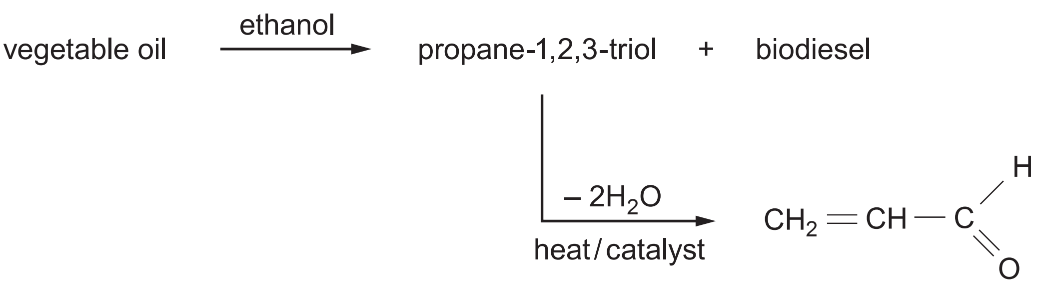
Examiner only

*(b)* Propenal, $CH_2{=}CH{-}CHO$ is made by two main methods.

**Method 1** From crude oil




**Method 2** From vegetable oil




(i) Suggest **one** reason why the general public would probably favour the production of propenal by Method 2. [1]

..................................................................................................................................................

..................................................................................................................................................

(ii) If you were considering which of these two methods to choose to produce propenal, state **two** other factors, apart from cost, that you would need to know to be able to come to a decision. [2]

Factor 1 ..................................................................................................................................

..................................................................................................................................................

Factor 2 ..................................................................................................................................

..................................................................................................................................................

Examiner only

(iii) The densities of biodiesel and propane-1,2,3-triol are shown in the table. These two liquids are immiscible (insoluble in each other).

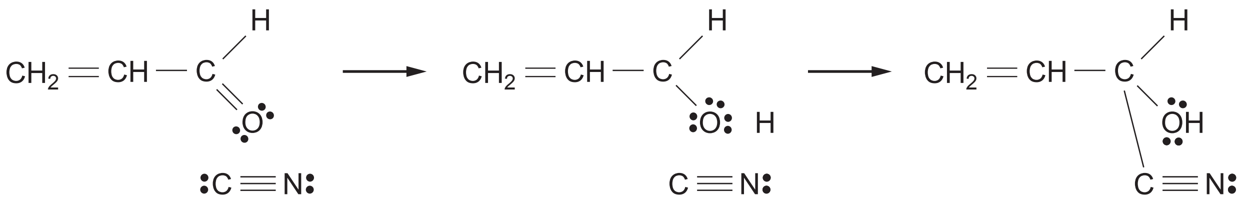
| Liquid | Density / $g\,cm^{-3}$ |
|---|---|
| biodiesel | 0.85 |
| propane-1,2,3-triol | 1.26 |

Sketch and name a piece of apparatus that could be used to separate a mixture containing $25\,cm^3$ of each liquid. The two liquids should also be labelled. [2]

Name of apparatus ..............................................................................................................

(iv) Propenal reacts with hydrogen cyanide by a nucleophilic addition reaction.

Complete the mechanism for this reaction using curly arrows and partial/complete charges as appropriate. [2]




1410U401 11

15